Year 11 Mathematics

Number 9

6.4 Reverse Percentages

Reverse Percentages

STARTER · 5 MIN

Date: _____ **Title:** Reverse Percentages

LEARNING OBJECTIVES

- Find the original amount after a percentage increase
- Find the original amount after a percentage decrease
- Solve reverse percentage problems, including VAT

STARTER *Answer in your book. No calculators.*

1. Write the multiplier for a 20% increase
2. Write the multiplier for a 30% decrease
3. Work out 60 ÷ 1.2
4. Work out 35 ÷ 0.7

Starter — Answers

STARTER

Tick or correct in green pen

1. **1.2**

2. **0.7**

3. $60 \div 1.2 = \mathbf{50}$

4. $35 \div 0.7 = \mathbf{50}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the original amount after a percentage increase	☐
SILVER	I can find the original amount after a percentage decrease	☐
GOLD	I can solve reverse percentage problems, including VAT	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Reverse Percentages

INTRODUCTION · 3 MIN

The original amount is always 100%. Work backwards from the new amount.

Original = 100%

The amount before the change.

After a 20% rise: 120%

Find 1%

Divide by the new percentage.

120% = 60, 1% = 0.5

Divide by the multiplier

New ÷ multiplier = original.

60 ÷ 1.2 = 50

Check

Apply the change to your answer.

50 × 1.2 = 60

Examples 1 and 2 — Using 1%

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

After a 20% increase, a price is £96. Find the original price.

Example 2

After a 25% decrease, a coat costs £60. Find the original price.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

120% = £96

1% = 96 ÷ 120

= 0.8

100% **= £80**

Tip: the new price is 120%, not 100%.

Example 2

75% = £60

1% = 60 ÷ 75

= 0.8

100% **= £80**

Tip: 100% − 25% = 75%.

Task 1

PRACTICE · 9 MIN

BRONZE *Find the original price*

1. After a 10% increase: £55
2. After a 50% increase: £90
3. After a 20% decrease: £40
4. After a 25% increase: £150
5. After a 40% decrease: £36

★ After a 5% increase: £84

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $55 \div 110 \times 100 = \mathbf{£50}$
2. $90 \div 150 \times 100 = \mathbf{£60}$
3. $40 \div 80 \times 100 = \mathbf{£50}$
4. $150 \div 125 \times 100 = \mathbf{£120}$
5. $36 \div 60 \times 100 = \mathbf{£60}$

★ $84 \div 105 \times 100 = \mathbf{£80}$

Examples 3 and 4 — Dividing by the Multiplier

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $\text{original} = \dfrac{\text{new amount}}{\text{multiplier}}$

Example 3

A price including 20% VAT is £234. Find the price before VAT.

Example 4

A car is worth £13 600 after losing 15% of its value. Find its original value.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Multiplier = 1.2

234 ÷ 1.2

£195

Tip: divide, do not take 20% off.

Example 4

Multiplier = 0.85

13 600 ÷ 0.85

£16 000

Tip: check: 16 000 × 0.85 = 13 600.

Task 2

PRACTICE · 9 MIN

SILVER

Find the original amount

Hint $\text{original} = \frac{\text{new amount}}{\text{multiplier}}$

1. After an 8% increase: £270
2. After a 12% decrease: £176
3. Including 20% VAT: £72
4. After a 35% decrease: £52
5. After a 2.5% increase: £451

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $270 \div 1.08 =$ **£250**
2. $176 \div 0.88 =$ **£200**
3. $72 \div 1.2 =$ **£60**
4. $52 \div 0.65 =$ **£80**
5. $451 \div 1.025 =$ **£440**

Examples 5 and 6 — Reverse Percentage Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint original = $\frac{\text{new amount}}{\text{multiplier}}$

Example 5

(3 marks)

A salary rises by 4% to £31 200. Find the increase in pounds.

Example 6

(3 marks)

Prices are cut by 30%, then by a further 10%. A TV now costs £378. Find the original price.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

31 200 ÷ 1.04 = £30 000

31 200 − 30 000

£1200

Tip: find the original first.

Example 6

0.7 × 0.9 = 0.63

378 ÷ 0.63

£600

Tip: multiply the multipliers.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint original = $\frac{\text{new amount}}{\text{multiplier}}$

1. A population increased by 15% to 9200. Find the original population. (2 marks)

2. A sale gives 22% off. The sale price is £117. How much is saved? (3 marks)

3. After 3 years of 5% compound interest an account has £2315.25. How much was invested? (3 marks)

4. A car loses 20% of its value, then 10% the next year. It is now worth £10 800. Find its original value. (3 marks)

5. After a 20% rise a price is £60. Ben says the original was £48. Explain his mistake and correct it. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $9200 \div 1.15 = \mathbf{8000}$

2. $117 \div 0.78 = £150, \quad 150 - 117 = \mathbf{£33}$

3. $2315.25 \div 1.05^3 = \mathbf{£2000}$

4. $0.8 \times 0.9 = 0.72, \quad 10\,800 \div 0.72 = \mathbf{£15\,000}$

5. He took 20% of the new price, $\quad 60 \div 1.2 = \mathbf{£50}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE After a 10% increase a price is £66. Find the original.

SILVER After a 25% decrease a price is £45. Find the original.

GOLD A price including 20% VAT is £150. Find the price before VAT.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $66 \div 1.1 = \mathbf{£60}$

SILVER $45 \div 0.75 = \mathbf{£60}$

GOLD $150 \div 1.2 = \mathbf{£125}$

Exam Practice — Using 1%

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1 (2 marks)

After a 20% increase, a price is £84.
Work out the original price.

..... £

Question 2 (3 marks)

In a sale, prices are reduced by 35%.
A dress costs £52 in the sale.
Work out the original price.

..... £

Exam Practice — Dividing by the Multiplier

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint original = $\frac{\text{new amount}}{\text{multiplier}}$

Question 3

(2 marks)

A price including 20% VAT is £186.
Work out the price before VAT.

..... £

Question 4

(3 marks)

A house is worth £261 000.
This is 8% less than last year.
Work out its value last year.

..... £

Exam Practice — Reverse Percentage Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint original = $\frac{\text{new amount}}{\text{multiplier}}$

Question 5

(3 marks)

After a 15% pay rise, Tom earns £33 350.

How much more does he earn now?

..... £

Question 6

(3 marks)

After 2 years of 4% compound interest, an account has £3785.60.

How much was invested?

..... £

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

120% = £84, 1% = £0.70 — **M1**

100% = £70 — **A1**

Tip: the new price is 120%.

Question 2

100% − 35% = 65% — **M1**

1% = 52 ÷ 65 = 0.8 — **M1**

100% = £80 — **A1**

Tip: check: 65% of £80 = £52.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

186 ÷ 1.2 — M1

= £155 — A1

Tip: divide, do not take 20% off.

Question 4

multiplier = 0.92 — M1

261 000 ÷ 0.92 — M1

= £283 695.65 — A1

Tip: check: 283 695.65 × 0.92 = 261 000.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

33 350 ÷ 1.15 = £29 000 — M1

33 350 − 29 000 — M1

= £4350 — A1

Tip: find the original first.

Question 6

multiplier = 1.04^2 — M1

3785.60 ÷ 1.0816 — M1

= £3500 — A1

Tip: divide by the multiplier to the power 2.